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INFS7205 ADVANCED TECHNIQUES FOR HIGH DIMENSIONAL DATA (ST LUCIA) - SEMESTER 1, 2026

Multi-Modal Agent for Social Media Biases

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1 Problem Framing [link](#) · [index](#)

This study is seeking to address the issue of the daily consumption of social media content, particularly the biases and the tendency to become immersed in YouTube videos that reinforce specific ideals. It is conceivable that an algorithm could be both the cause and the solution, by making users aware of the beliefs to which they are exposed. Therefore, our research questions is the following:

Is it possible to design a personalized multi-modal assistant that performs better than an out-of-the-box LLM agent at searching for, identifying, and communicating any existing biases in a YouTube watch history?

We need an *agent* in the sense of a **MRAG-powered LLM model** to address the multi-modality (i.e., titles and thumbnails) and personalization parts. This specific technology is chosen due to the ambiguity of defining *measurable* levels of "bias" and "rabbit hole" while taking advance of the *dynamic* nature of LLM in processing and comprehending the user's watch history. Such dynamicity, however, creates the additional need of *rigorously* verifying the contribution of each component to the overall performance.

2 Retrieval-Augmented Generation (RAG) [link](#) · [index](#)

Among the different multi-modality strategies [1], in this study we decided to go for *strategy 3*, consisting of a **multi-modal knowledge base and a mono-modal LLM**. This means that the LLM itself is *natively* only able to process text, but it acquires its multi-modal capabilities through retrieval augmentation from the knowledge base, whose vector storage allows the agent to (indirectly) reason about different modalities. This design decision is rooted in the ambition to address the task without immediately resorting to the most sophisticated (and costly) model available.

2.1 Dataset and augmentation strategy

The dataset we decided to use [2] originally contains 1.844.919 watch history entries¹ belonging to 244 different anonymous users in a time span from the 2010-08-08 01:44:39 to the 2018-09-26 23:03:33.

However, due to the limits in 3.1, we had the need to narrow down to 987 rows belonging to the user 25² only. While this heavily narrows the generalization of the solution, it was a necessary optimization step for an architecture that can be easily expanded to different users.

We created two custom scripts [3], where the first retrieves the video titles by means of the YouTube APIs [4], and the second downloads in parallel all the static thumbnails from an encoded URL.

2.2 Tokenization

A prerequisite for the embedding model is the tokenization of all corpus supplied to it. We experimented [3] with Jinaa and CLIP [5] tokenizers and embedding models for this purpose. The first provided slightly better semantic results (particularly for shorter queries) but suffered in turn from a heavy computational cost, therefore we opted for the CLIP model³.

¹Each row contained the following columns: `log_id`, `user_id`, `video_id`, `watch_date`, `subscribed`, `playlist_name`

²Such user was one of those having the longest watch history

³`sentence-transformers-clip-ViT-B-32` with $d = 512$.

2.3 Vector database

Among the tested database platforms we tried FAISS [6] and ChromaDB [7]. While the first provided a fast and lightweight storage, its temporary nature (being limited to RAM memory) made us opt for the ChromaDB in order to reduce the startup time.

2.4 Indexing

Even though text and images had to be embedded separately, we soon realized they semantically represented the same meaning of interest (according to the problem definition 1). Therefore, we decided to linearly combine their normalized embeddings, giving more weight to the title (as assumed less noisy than the image content). For example:

$$e_{\text{fusion}} = 0.7 \cdot \frac{e_{\text{text}}}{\|e_{\text{text}}\|} + 0.3 \cdot \frac{e_{\text{img}}}{\|e_{\text{img}}\|} = 0.7 \cdot \begin{pmatrix} 0.12 & 0.07 & \cdots & 0.05 \end{pmatrix} + 0.3 \cdot \begin{pmatrix} 0.02 & 0.11 & \cdots & 0.04 \end{pmatrix} \in \mathbb{R}^{512} \quad (1)$$

The overall time required to tokenize, embed and store all the records mentioned in section 2.1 can be found in table 2:

3 Agent Pipeline Design [link](#) · [index](#)

The whole agent infrastructure is managed from the popular framework LangGraph [8], which provides a convenient platform to wrap and connect together the LLMs, routers, tools and memory presented in the following sections.

3.1 LLM

The core of our agent pipeline is certainly the LLM model that powers each of the agents. It receives the messages history from the framework [8], reasons about whether to call a tool, think or respond, and then returns this decision.

Experimenting with *many* different models³ and API providers, we discovered a *large* difference in their respective ability to understand instructions and invoke the correct tool with the correct arguments. We eventually opted for the `gpt-4o-mini` as the main model.

3.2 Tools

The model is provided with the following tools:

- `find_similar_videos`: queries the vector database 2.3 for semantically similar videos by tile or thumbnail
- `retrieve_session`: retrieves a subset of the dataset 2.1 containing the watch history of the user
- `analyze_bias_profile`: invokes the subagent 3.4 to produce the bias profile of the previously retrieved session

3.3 Memory

Other than the chat history, (through each tool 3.2) call is enriched with the following fields:

- `watch_history`: list of items from the previous history retrieval

³openai/gpt-oss-120b, gpt-4o-mini, qwen3:0.6b-q4_K_M, llama-3.3-70b-versatile, among the others

- **user_id**: tells the `retrieve_session` tool which user to look for
- **thread_id**: randomly generated UUID that allows the model to remember previous interactions

3.4 Subagents

An interesting topic documented [8] is the use of sub-agents, with one of the simplest implementations being a tool call to wrap another agent’s invocation (`analyze_bias_profile` from 3.2). To improve this component’s performance, we applied a `response_format` constraint over the following fields: `emotional_tone`, `sensationalism`, `evidence_titles`, `overall_profile`, `confidence` and others. The last three properties, together with the system prompt A.8, help identify hallucinations.

4 Evaluation [link](#) · [index](#)

4.1 Ablations

The model, since made up of different components, needs to be tested in the effects of the introduction or removal of each of them. To this goal, we identified the following set of architectures to test:

Ablation	Short-Term Memory ²	KB	Vector DB Indexing	Bias Analysis Sub-Agent
0. Vanilla	Y	N/A	N/A	N
1. Text RAG	Y	CSV	Text	N
2. MM-RAG	Y	CSV + Thumbnails	Hybrid	N
3. Agentic MM-RAG ¹	Y	CSV + Thumbnails	Hybrid	Y

Table 1: Ablation study configurations testing different components of the agent system.

4.2 Augmentation

Since the dataset contains general-purpose watch history [2], we augmented it with synthetic rows introducing controlled biases to enable meaningful downstream predictions. To this end, we created an additional agent¹ to generate biased titles and plausible watch dates. IDs were assigned using the first search result `ytdlp` for each title, enabling retrieval of coherent thumbnails. After validation, the new rows were merged with the original dataset and shuffled.

This resembles a teacher-student setup, where the tester agent (2) was intentionally made more capable (`openai/gpt-oss-12` with `reasoning_effort="medium"`).

4.3 Q/A generation

To generate question-answer pairs for the augmented test set, we created a new agent¹ with the ablation structure from 1, dataset sampling tools, and a specialized sub-agent (prompt in section A.8) to propose Q/A pairs. It was embedded in a routing-based graph, allowing iterative dataset exploration until stopping.

This process required substantial human intervention to ensure the Q/A pairs remained grounded and non-overlapping. The final set is reported in 3.

4.4 Answers scoring

Based on the chosen evaluation criteria, we identify three categories:

¹This agent is the one presented in all the sections so far.

²The effect on providing or not the message history was not considered of interest for the study.

1. **Correctness:** factual alignment with the expected answer.
2. **Faithfulness:** grounding in the provided context, avoiding hallucinations.
3. **Usefulness:** relevance and practical completeness.

We focused on correctness using the LLM-as-a-Judge technique. As in the previous section, we created an additional agent¹ provided with the questions, golden answers (4.3), and model outputs from 1, and tasked it with scoring each response.

Overall, *model_1* and *model_2* perform most consistently, *model_3* is the most uneven, and *model_0* is highly task-dependent, excelling on factual questions but performing poorly on conversational ones. All models struggle with cross-modal questions. Score distributions are available in A.6.

4.5 Efficiency

To further compare the four ablations, we also measured token usage across questions³. The resulting distributions are shown below, together with the cumulative token trend across question progression.

The plots suggest an efficiency-stability tradeoff: *model_0* is highly efficient on factual questions but unstable elsewhere, *model_1* and *model_2* are the most balanced with steadier token growth, while *model_3* is the least consistent and generally the most expensive, especially on analytical tasks. Efficiency distributions are available in A.7.

5 Wrapping Up [link](#) · [index](#)

5.1 Failure analysis

The main limitation was model availability. Free tiers of the API platforms imposed strict constraints, forcing choices such as batch testing, a single-user setup, and frequent model switching, reducing rigor and generalizability. Models also often failed silently, making it hard to separate framework, implementation, and model limitations. A paid OpenAI plan was ultimately required.

The scope in 1 also limited improvements to the tokenization strategy 2.2, resulting in a poorly structured semantic space where generic videos (e.g. "SkiVel") could rank highly regardless of the query.

5.2 Success analysis

We adopted a hands-on, human-first approach to maximize learning, making the full development process publicly available [3]. The structured testing methodology also provided a solid basis for evaluation 4.

5.3 Conclusions

We presented a personalized multi-modal agent system 2 that retrieves and analyzes YouTube watch history 1 through a fused representation and specialized sub-agents 3. The performance and efficiency was evaluated in 4 through a custom LLM-as-a-Judge setup against meaningful ablations.

¹This model's prompt can be found in section A.8

²For brevity, we omit the exact invocation details. Given the number of evaluations ($4 \cdot 4 \cdot 4 = 64$), factual questions were batched, while conversational ones included one prior question-answer pair for context.

³Since each category within each model reused the same thread ID, token counts are cumulative, except for factual questions, which were grouped separately to limit excessive usage.

A Appendix [link](#) · [index](#)

A.1 Overall setup

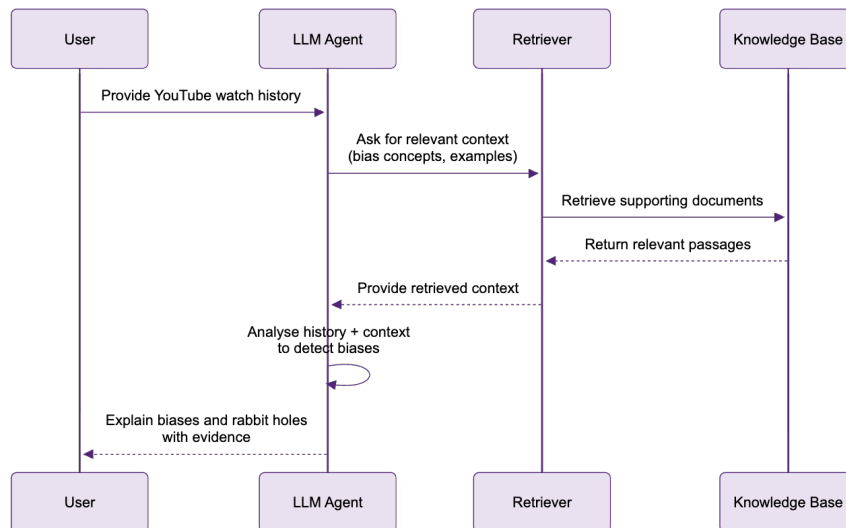


Figure 1: high-level sequence of the proposed agent's steps

A.2 Overall testing setup

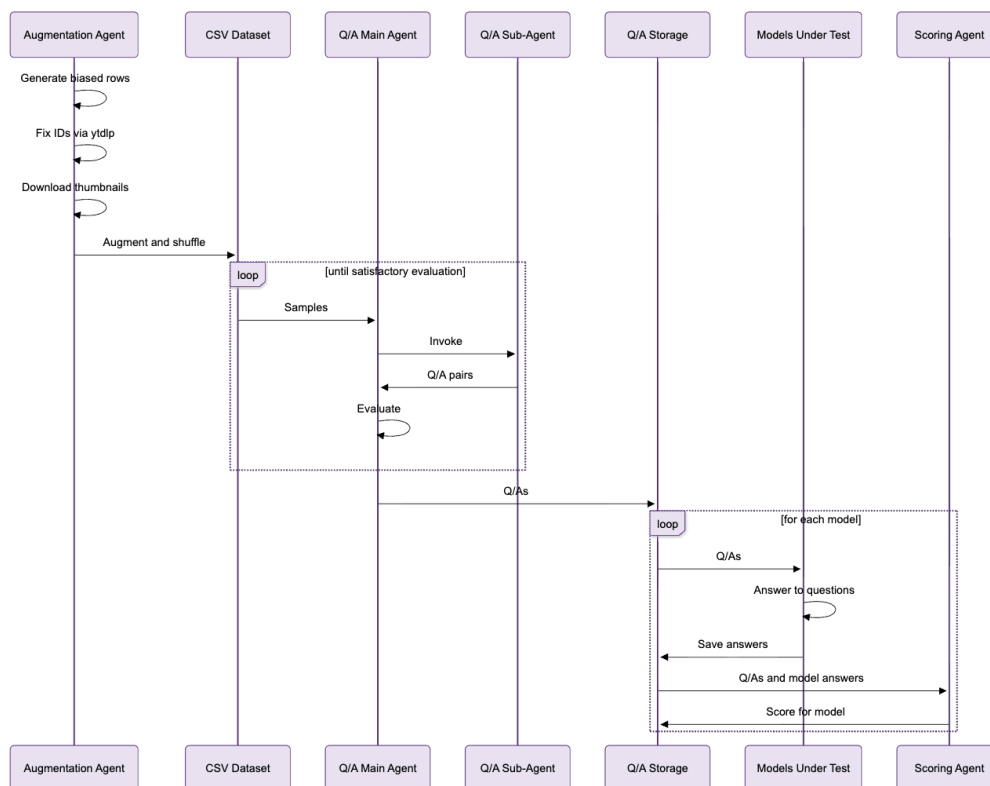


Figure 2: High-level sequence of the evaluation pipeline.

A.3 Dataset embedding time

Multi-modality setup (Table 1)	No. of rows	Total embedding time
Text only	1173	31 seconds
Hybrid	1173	1 minute and 45 seconds

Table 2: Indexing time of test set 4.2 by multi-modality type.

A.4 Web interface

While the first conversational attempts were carried out using a terminal approach, we created a custom human-refined web UI where it's possible to communicate with the agent. The interface allows to attach a single document (restricted not due to model limitations but just for testing purposes) and a message, then the back-end server running with **FastAPI** **fast-api** forwards the request to agent as usual. Once the agent responds, the answer is presented in the chat. Easiness of use, cleanliness of the interface and deep knowledge of the underlying code was a priority here just like in the rest of the code. It's possible to run the interface through the publicly available code [3].

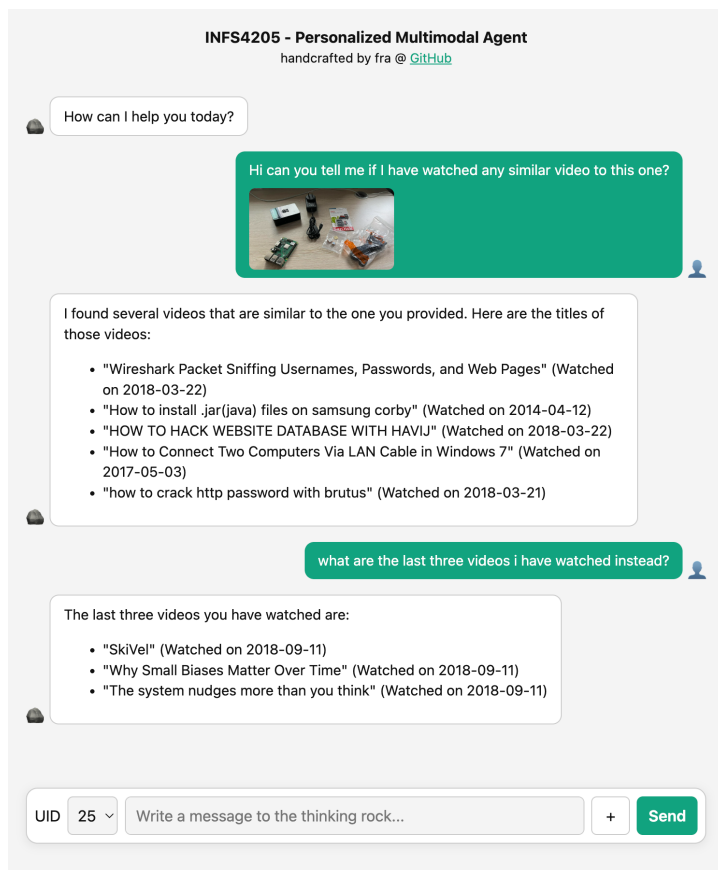


Figure 3: Multi-modal conversation in the web interface. Even though the model doesn't natively have multi-modal capabilities, it correctly matched IoT-networking content related to the picture by means of the vector database ?? similarity search

A.5 Question and Answers

Factual

1 **Q:** What is the exact watch date and time for the video titled “Why Small Biases Matter Over Time”?
A: 2018-09-11 16:59:59

2 **Q:** Which video ID corresponds to the video “SkiVeI”?
A: F8wJdvXK5yU

3 **Q:** What is the title of the video watched on 2018-09-11 at 15:42:17?
A: Why small biases matter

4 **Q:** Which title has the video with ID 8DUSSbJPfwU?
A: The system nudges more than you think

Cross-modal

5 **Q:** Based on the thumbnail at ../knowledge/thumbnails/z47r66-9XIg.jpg, which video ID of the user’s watch history is the most similar?
A: 6CO6ShQd5i0

6 **Q:** Based on the thumbnail at ../knowledge/thumbnails/z47r66-9XIg.jpg, which video title of the user’s watch history is the most similar?
A: Sniper 3D Assassin Al Vahdeko Spec Ops Mission 1- EVERYBODY HATES JIM

7 **Q:** Based on the title “Why liberals keep missing this point”, which video ID of the user’s watch history is the most similar?
A: B5fq2j6Z9ro

8 **Q:** Based on the title “Why liberals keep missing this point”, which video title of the user’s watch history is the most similar?
A: Why Liberals Destroy Culture

Analytical

9 **Q:** What recurring topics does the user’s watch history reveal based on the most recently watched 50 videos?
A: Two recurring themes: (1) kids cartoons and nursery rhymes (Ben 10, StoryBots, Wiggles, Disney Kids), and (2) media/cognition/bias topics (The Subtle Effects of Repetition, How Exposure Shapes Beliefs, Why People Gravitate Toward Familiar Ideas, Why small biases matter).

10 **Q:** What polarizing patterns, if any, does the user’s watch history reveal based on the least recently watched 50 videos?
A: Repeated anti-liberal framing in titles, e.g., “Why Liberals Destroy Everything Good”, “Why Liberals Hate Tradition”, “Why liberals keep missing this point”, “Why Liberals Destroy Culture”, and “Why Liberals Are Destroying Tradition”.

11 **Q:** What political biases, if any, does the user’s watch history reveal based on the the most recently watched 50 videos?
A: No clear partisan political bias. The most recent 50 are dominated by kids cartoons/children’s songs and general media/cognition commentary (e.g., The Role of Framing in Everyday News, Why Content Feeds Reinforce Views, The Gradual Shift in Online Discourse, Why Small Biases Matter Over Time).

12 **Q:** Which health-related misinformation topics, if any, does the user’s watch history reveal based on the least recently watched 50 videos?

A: Vaccine misinformation, e.g., “The Vaccine Hoax They Don’t Want You to Know” and “The Vaccine Truth Finally Revealed”.

Conversational

13 **Q:** Using a tool, which title is the most similar to that?

A: The Vaccine Hoax They Don’t Want You to Know.

14 **Q:** What is its video ID?

A: CTj_xoCuhPU

15 **Q:** Using a tool, can you find the title of a similar watched video?

A: The Vaccine Truth Finally Revealed

16 **Q:** Do you think those videos present misinformation?

A: Yes: “The Vaccine Hoax They Don’t Want You to Know” and “The Vaccine Truth Finally Revealed” use common conspiracy/misinformation framing (e.g., “hoax”, “truth finally revealed”).

Table 3: Evaluation questions and golden answers grouped by category.

A.6 Model ablations scoring

Category	Q	Model 0	Model 1	Model 2	Model 3
Factual	1	1.0	0.7	0.7	0.7
	2	1.0	0.8	0.8	0.8
	3	1.0	0.8	0.8	0.7
	4	1.0	0.7	0.7	0.7
	<i>Avg</i>	<i>1.00</i>	<i>0.75</i>	<i>0.75</i>	<i>0.73</i>
Cross-modal	5	0.0	0.0	0.0	0.7
	6	0.0	0.0	0.3	0.7
	7	0.0	0.0	0.0	0.7
	8	0.0	0.0	0.3	0.7
	<i>Avg</i>	<i>0.00</i>	<i>0.00</i>	<i>0.15</i>	<i>0.70</i>
Analytical	9	0.6	0.6	0.6	0.6
	10	0.7	0.9	0.9	0.7
	11	0.9	0.9	0.7	0.0
	12	0.9	0.9	0.9	1.0
	<i>Avg</i>	<i>0.78</i>	<i>0.83</i>	<i>0.78</i>	<i>0.58</i>
Conversational	13	0.0	1.0	1.0	1.0
	14	0.0	1.0	1.0	1.0
	15	0.0	1.0	1.0	1.0
	16	0.7	0.7	0.7	0.7
	<i>Avg</i>	<i>0.18</i>	<i>0.93</i>	<i>0.93</i>	<i>0.93</i>
Average		0.49	0.63	0.65	0.73

Table 4: Evaluation scores across 4 model variants with category and grand averages.

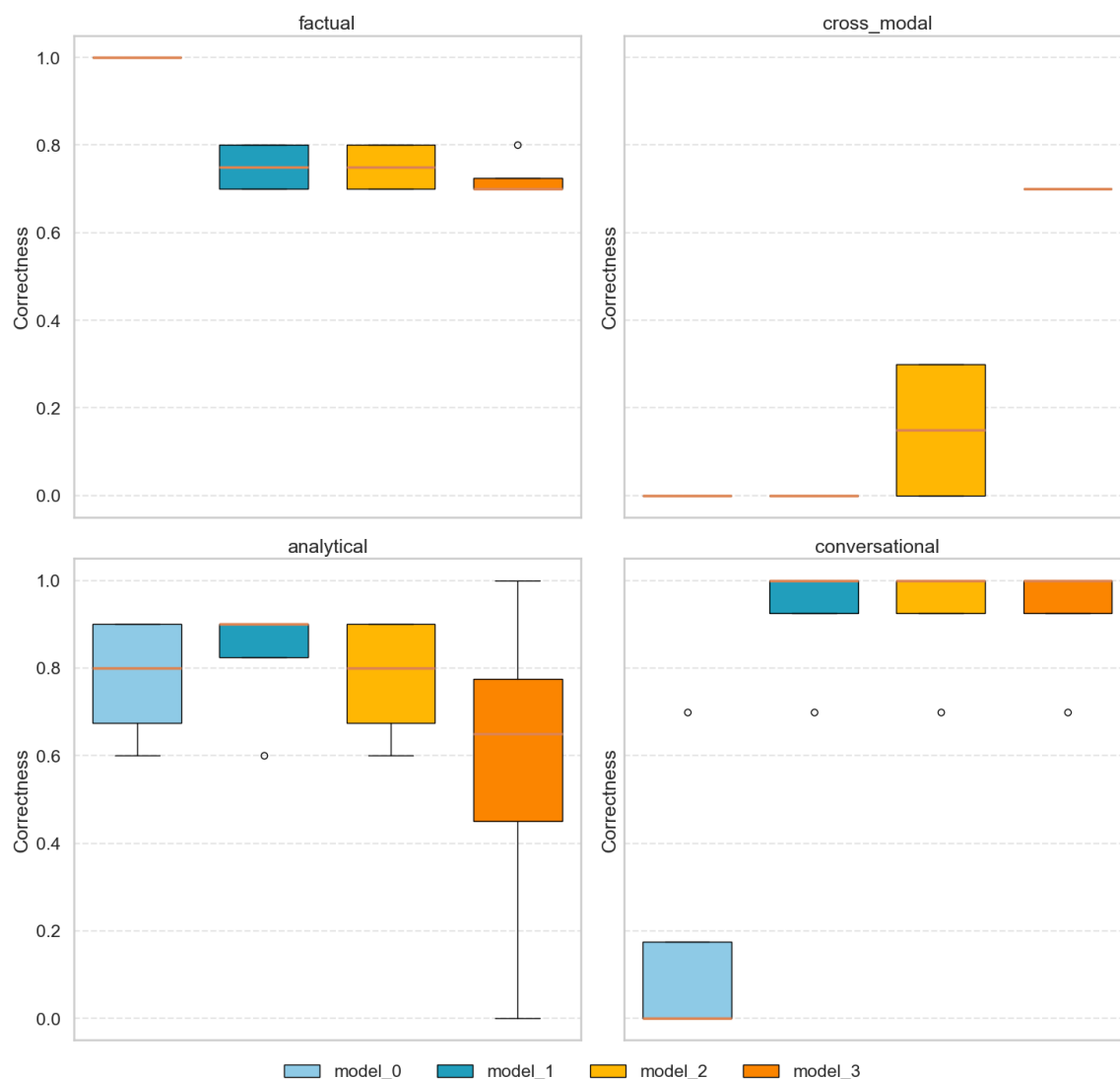


Figure 4: Distribution of the LLM-judge scores assigned to each ablation.

A.7 Model ablations efficiency

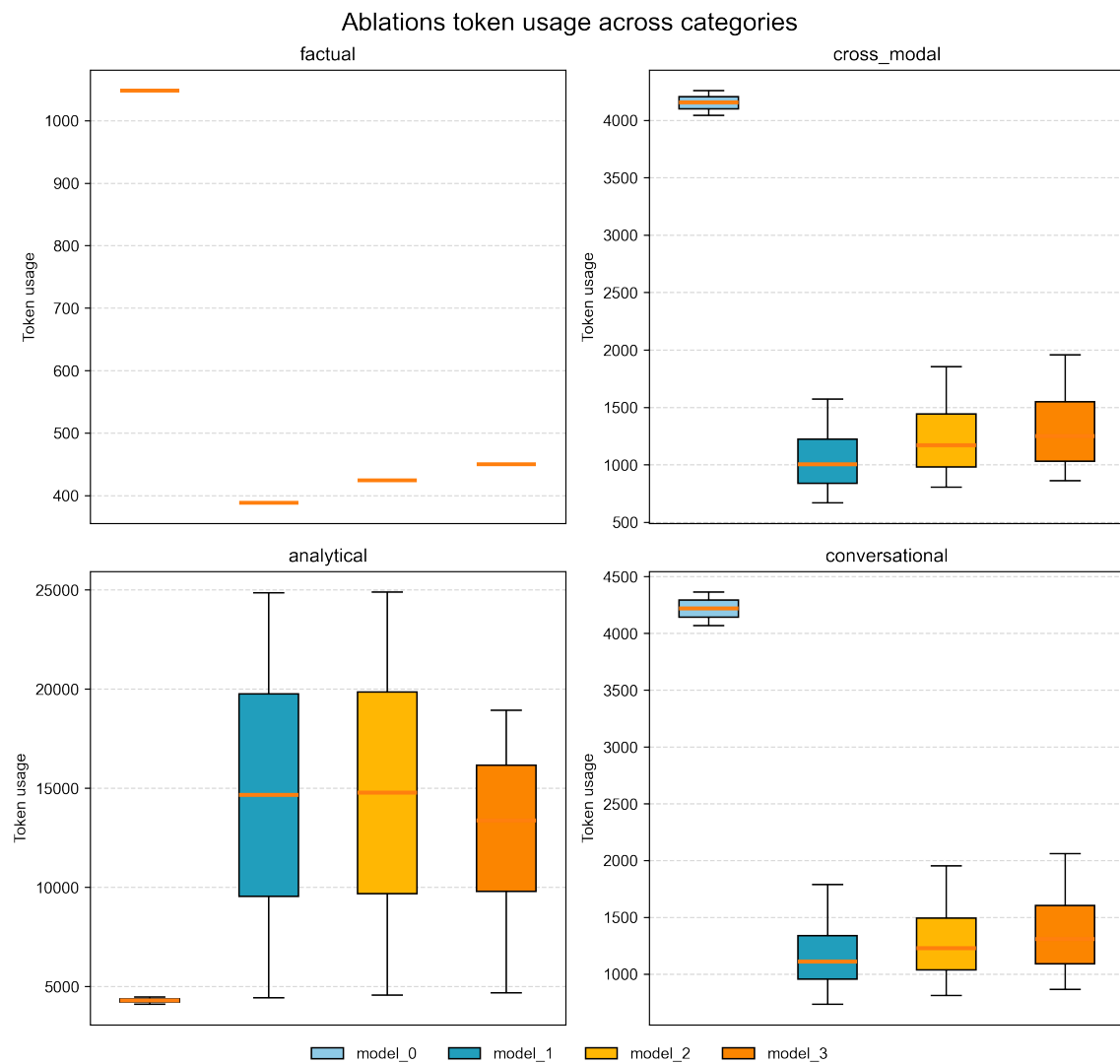


Figure 5: Trend of the token usage from each ablation.

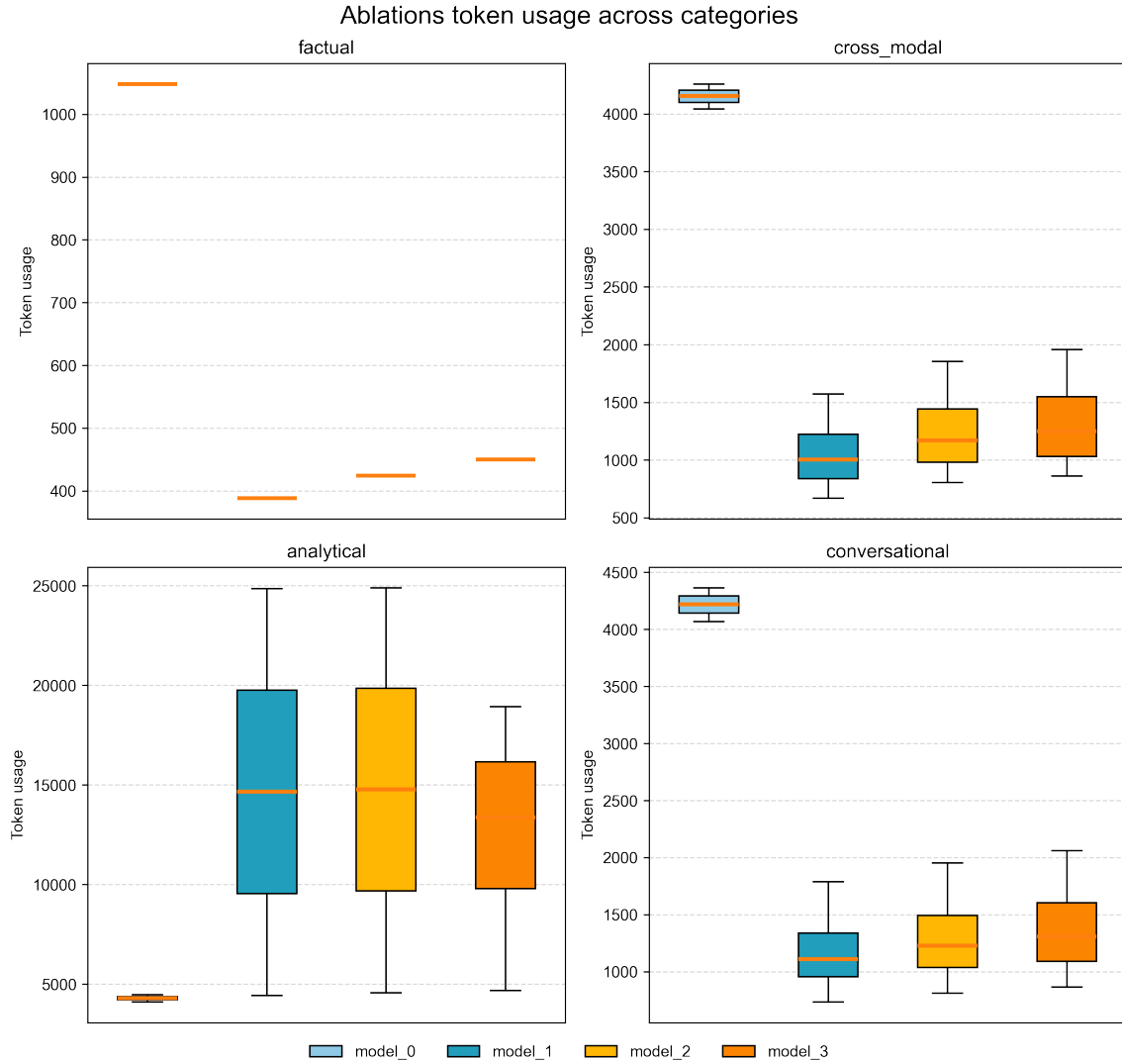


Figure 6: Distribution of the per-question token usage from each ablation.

A.8 System prompts

A.8.1 Main agent

You are a YouTube watch-history analysis agent focused on bias, polarization, sensationalism, clickbait, emotional tone, and
 ↪ rabbit-hole effects.

Rules:

- If you are sure about the user's intent AND it matches a tool's purpose, call that tool with appropriately typed arguments.
 ↪ Otherwise, do not call any tool and ask for clarification.
- When asked about bias, polarization, sensationalism, clickbait, emotional tone, or rabbit-hole effects, ensure the watch
 ↪ history was retrieved using the relative tool.
- Base all claims on factual data from watch history and tool results. For each claim, mention a specific example from the
 ↪ watch history that supports it (e.g., a video title or topic). Avoid vague generalizations.
- Be cautious and do not overclaim ideology; if unsure, say that the evidence is weak or ambiguous and optionally ask for
 ↪ clarification.
- When given an image, use the find_similar_videos tool immediately, passing the image's local file path directly to the tool.
- When asked to display an image, respond with the [markdown tag](URL), where URL is the best image URL from user input, state,
 ↪ memory, or tools.
- Do not use formatting (e.g. **bold**, *italic* and so on). Do not print <think> in your answers. If answering multiple
 ↪ questions, split each answer in a new row with a leading "*".

- Try to be as concise as possible.

A.8.2 Bias analysis subagent

You analyze a single YouTube watch session and return a structured bias profile and assessment of rabbit-hole effects.

```
# Rules
- Use only the provided watch-session records; do not assume any additional history.
- If no watch history is provided, explicitly refuse to analyze.
- Be cautious and conservative in your conclusions.
- Do not overclaim ideology; avoid assigning strong labels without clear evidence.
- Focus your analysis on:
  - Ideological or political lean (if any).
  - Topic concentration versus diversity.
  - Emotional tone and sensationalism.
  - Escalation or narrowing over time (rabbit-hole behavior).
```

A.8.3 Test dataset augmentation agent

Generate 50 rows to augment an existing YouTube watch log dataset, introducing detectable biases in content consumption. Each row should represent a video watched by a user with varied timestamps, realistic thumbnail URLs, and durations between ↪ 5-45 minutes.

```
# Bias Categories & Title Examples:
- Rightist Extremism: "Why Liberals Destroy Everything", "The War on Western Civilization", "Globalist Agenda EXPOSED" and
↪ similar
- Leftist Extremism: "Why Capitalism Must End", "Fascism Returns: The MAGA Threat", "Corporate Greed is Killing Us" and similar
- Conspiracy (sensational framing): "COVID Vaccine Hoax Finally Revealed", "Election 2020: The Stolen Truth", "Climate Change:
↪ The Biggest Scam" and similar
- Sensational/Clickbait: "SHOCKING: What They Hide From You", "You Won't Believe This Election Fraud", "EXPOSED: The 5G Danger
↪ Nobody Talks About" and similar
- Polarization: "Us vs Them: The Culture War", "Red vs Blue: America Divided", "The Great Divide: Why We Can't Unite" and
↪ similar
- and more...
```

```
# Output Format:
log_id,user_id,video_id,watch_date,subscribed,playlist_name,video_title
```

```
# Rules:
- log_id = unique random integer with 6 digits
- user_id = always 25 fixed
- video_id = unique random string of 11 characters
- watch_date = realistic timestamps between 2014-04-12 06:38:03 and 2018-09-11 17:02:26
- subscribed = always None
- playlist_name = always None
- video_title = realistic video titles referring to one or more of the above biases (IMPORTANT)
```

```
# Example row:
146533,25,Q5JnbpltAAo,2018-09-10 22:59:44,None,None,The Evil Jungle | Ben 10 | Cartoon Network
```

Give me a final recap of what you generated, with a breakdown of some of the biased rows produced.

A.8.4 Test question-making main agent

You are an expert evaluator for multimodal agent systems. Your goal is to generate specific test queries for a YouTube watch history bias analysis agent.

You have already inspected samples from the dataset using tools. Use ONLY evidence observed in those samples and the tool ↪ outputs in the conversation.

```
## Query Categories
Generate exactly 4 questions for each category:
```

1. Factual retrieval: direct lookup from watch history CSV
2. Cross-modal retrieval: given a title or thumbnail URL, find a similar video in the history
3. Analytical multi-hop synthesis: detect a bias pattern across multiple videos
4. Conversational follow-up: memory-dependent queries referencing prior responses

Architectures to benchmark

Your questions must help differentiate:

1. Vanilla: no KB / retrieval
2. Text RAG: CSV + text-only indexing
3. MM-RAG: CSV + thumbnails + hybrid indexing
4. Agentic: MM-RAG + bias analysis sub-agent

Rules

- Return exactly 4 items per category.
- Every question must have a grounded golden answer.
- Golden answers must be based on rows actually observed through the tools.
- Conversational questions must refer to earlier generated questions or answers.
- Cross-modal questions must clearly require thumbnail/title similarity.
- Analytical questions must test meaningful bias patterns in the data.
- Avoid vague or generic questions.
- Do not invent entities not supported by the observed samples.

A.8.5 Test question-making sub-agent

You are generating benchmark questions for a YouTube bias-analysis agent.

You may call tools repeatedly to inspect different random splits of the dataset.

Before stopping, gather enough evidence to support:

- 4 factual questions
- 4 cross-modal questions
- 4 analytical questions
- 4 conversational follow-up questions

You should inspect multiple random samples and look for:

- exact titles, timestamps, and repeated channels
- thumbnail/theme patterns for cross-modal similarity
- detectable bias trends such as political leaning, sensationalism, conspiracy framing
- candidate question-answer pairs grounded in observed rows

Do not produce the final JSON yet.

When you have enough evidence, say clearly that you are ready for final structuring.

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